Arpan Dey

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Skills

- **Languages:** C, C++, JavaScript (ES6+), Python, Java, Go
- **Frameworks/Libraries:** Express.js, ReactJS, Spring Boot, Matplotlib, Socket.IO, Flask
- **Databases:** MongoDB, MySQL
- **Tools:** Docker, Jenkins, Git/GitHub, Postman
- **Soft Skills**: Resilient, Resourceful, Adaptable, Responsible

Projects

- **Kanvex – Project Collaboration Tool** *Sept' 2025 - Nov' 2025*
  - Built a real-time workspace enabling teams to plan, track, and collaborate on tasks efficiently.
  - Developed role-based access, authentication, CRUD flows for boards and state sync across sessions using REST APIs.
  - Reduced task operations latency by 30-40% through efficient query batching, resulting in smooth, low-latency interactions for users.
  - **Tech Stack:** MongoDB, Express.js, ReactJS, Node.js
- **Low-Bandwidth Chat-App** *July 2025*
  - Created a communication system optimized for unstable or limited network environments using lossless text compression.
  - Implemented real-time communication using Flask with Socket.IO, a lightweight HTML/JavaScript interface, and custom Huffman encoding/decoding for efficient text transmission.
  - Achieved 35–55% data reduction per message, enabling reliable and efficient messaging in low bandwidth scenarios.
  - **Tech Stack:** Python, Socket.IO, Flask
- **Page Replacement Simulator** *Mar' 2025*
  - Engineered a visual simulator to compare page replacement algorithms for easier conceptual understanding of memory management.
  - Built GUI using Tkinter for dynamic input supporting 50+ page references, visualization and step-by-step tracing and added Matplotlib graphs for fault comparisons.
  - Enhanced understanding of FIFO, LRU and Optimal page replacement algorithms through an interactive, intuitive visualization tool.
  - **Tech Stack:** Python, Tkinter, Matplotlib

Certificates

- CI/CD Implementation for DevOps | Infosys Springboard *June 2025*
- Digital Systems: From Logic Gates to Processors | UAB Barcelona | Coursera *Oct' 2024*
- Introduction to Hardware and Operating Systems | IBM | Coursera *Sep' 2024*

Achievements

- Secured International Rank 111 in 10th SOF International Mathematics Olympiad.
- Ranked Top 3 in Google Cloud Capture the Flag event.
- Maintained a 700+ day streak and solved 1400+ problems across GeeksforGeeks and LeetCode.

Extra-Curricular Activities

- Managed event promotions and boosted campus engagement as **Coding Ninja's Campus Ambassador**.
- Led community initiatives including plantation drives, blood donation events and digital campaigns as **Campus Ambassador of MyGovIndia**.
- Participated in peer learning, tech events and collaborative projects via **Google Developer Students Club**.

Education

- **Lovely Professional University** Phagwara, Punjab
  Bachelor of Technology - Computer Science and Engineering: ***CGPA: 8.57*** *Aug' 2023 - Present*
- **Delhi Public School** Ranchi, Jharkhand
  Intermediate: ***Percentage: 88.00%*** *June 2021 - May 2022*
- **TPS DAV Public School** Baharagora, Jharkhand
  Matriculation: ***Percentage: 92.25%*** *Apr' 2019 – Mar' 2020*